

## 研究计划

### STUDY PLAN

研究计划是申请材料中的重要组成部分，通过陈述你的学习背景、学术兴趣等方面，能帮助学校更好的了解你。研究计划应不少于 **1,000** 字，并用**申请授课语言书写（中文或英文）**。  
This study plan is an important part of your application, which will help us to know you better. The study plan should be no less than **1,000** words in length and should be written **in the language of instruction you are to apply (English)**

|                                               |                                                                                                         |                                  |                                                             |
|-----------------------------------------------|---------------------------------------------------------------------------------------------------------|----------------------------------|-------------------------------------------------------------|
| 姓<br>FAMILY NAME                              | Momini                                                                                                  | 名<br>GIVEN NAME                  | Muhammad Saleem                                             |
| 性别<br>GENDER                                  | Male                                                                                                    | 国籍<br>NATIONALITY                | Pakistan                                                    |
| 出生日期<br>DATE OF BIRTH                         | 07/05/1994                                                                                              | 护照号<br>PASSPORT NO.              | NC5757661                                                   |
| 申请层次<br>DEGREE APPLIED                        | Ph.D(Physics)                                                                                           | 授课语言<br>INSTRUCTIONL<br>LANGUAGE | English                                                     |
| 申请院系<br>COLLEGE<br>/SCHOOL APPLIED            | School Of Physics &<br>Optoelectronic Engineering                                                       | 专业<br>PROGRAM<br>APPLIED         | Optical Engineering                                         |
| 导师姓名<br>NAME OF SUPERVISOR                    | Ouyang Qiuyun 欧阳秋云                                                                                      | 毕业院校<br>GRADUATED<br>FROM        | University of Management<br>& Technology Lahore<br>Pakistan |
| 申请研究方向名称/<br>EXPECTED RESEARCH<br>FIELD/TOPIC | Advanced Nonlinear Optical Phenomena in Nanostructure Materials for Next<br>Generation Photonic Devices |                                  |                                                             |

### 教育背景

#### EDUCATIONAL BACKGROUND

|   | From<br>(开始<br>时间) | To<br>(结束<br>时间) | Full School Name<br>(就读学校全称)                                   | Degree<br>Level<br>(学历) | Major<br>(专业) | Your<br>Average<br>Score<br>(GPA) | Full Score<br>(总分) |
|---|--------------------|------------------|----------------------------------------------------------------|-------------------------|---------------|-----------------------------------|--------------------|
| 1 | 2019.9             | 2021.11          | University of<br>Management &<br>Technology Lahore<br>Pakistan | MS(18year)              | Physics       | 3.18                              | 4                  |
| 2 | 2017.1             | 2019.4           | University of<br>Karachi, Karachi<br>Pakistan                  | M.sc<br>(16year)        | Physics       | 2.8                               | 4                  |

## 学术背景及成果 Academic Background and Achievement

(学术背景可包括过去学习经历、和与申请专业领域相关的工作和项目经历；学术成果可包括已发表的论文、专利、奖项等。Academic background refers to your previous education, and/or work and project experience related to the applied program; academic achievement refers to published papers, patents, awards, etc.)

### Academic Background and Achievement:

I completed my MS (Eighteen year education) in Physics in September 2021 from the **University of Management and Technology Lahore**, Pakistan. I have always been a fast learner and have consistently felt motivated to gain new knowledge from insightful individuals and collaborative teams. Which help me to complete my MS with one publication. Then University of Management and Technology helped me in the improvement of my knowledge to great extent. I involved myself in laboratory work, where I learned how to handle different apparatus and software.

Throughout my graduate and undergraduate studies, I come to know the wide-ranging practical applications of Physics. This sparked my curiosity and ignited a deep desire to learn, driving me to pursue my chosen courses. Core subjects such as **experimental techniques** in physics, solid-state physics, advanced **quantum mechanics**, advanced classical mechanics, advanced mathematical methods for scientists and engineers, advanced electrodynamics, **optoelectronics**, plasma physics, and many others have provided me with a comprehensive and profound understanding of physics. Based on this knowledge, I have researched in the field of material science and wrote my thesis under the supervision of Associate Professor **Dr Qayyum Zafar**, MY research topic was **“Integrated Capacitive and Resistive type Bi-modal Relative Humidity Sensor Based on 5,10,15,20-tetraphenylporphyrinatonicel (II) (TPPNi)–Zinc Oxide (ZnO) nanocomposite”** and I have also published **an article of the same title** from my thesis in American chemical society (ACS Omega). For this publication, I got a cash prize of “UMT research reward” of 50000 from the University of Management and Technology Lahore. I also got a “prime minister laptop” during my undergraduate study in 2018.

During my master’s studies after my course work, I have synthesized organic as well as inorganic materials for the application of humidity sensors and optoelectronic devices. I also studied and characterized their photophysical properties, crystal structure, and surface morphology using UV-visible spectroscopy, X-ray diffraction (XRD) and scanning electron microscopy (SEM) respectively. Furthermore, I also worked on origin software to draw graphs and used Chemdraw to draw the diagram of the experimental setup.

However, I am very interested and want to do some research in the field of **physics (modern photonic technologies)**. I find this field highly fascinating and approachable, making it both enjoyable and effortless to learn and work in. In particular, I am passionate about working on nonlinear optoelectronic devices, organic and inorganic semiconductor-based sensors, perovskite solar cells, and exploring the photo-catalytic properties of nano-materials.

In addition to academic studies, life has taught me to draw valuable lessons from everyday experiences and practical situations. My field of study has also made my life more intellectually vibrant and expansive. During my undergraduate and master’s studies, I was an active and engaged student, often participating in various curricular and co-curricular activities, such as attending conferences, seminars, and workshops.

A social and hardworking, who excels in working in challenging environment and who is always consistent in achieving the best results by solving technical and social issues, analyzing information

and digesting figures. I am a calm and introspective person who enjoys a peaceful lifestyle and has a strong ability to focus. I am well-organized, hardworking, and dedicated to steadily pursuing my goals. When I commit to a task, I am determined to see it through to completion. My years of living in a hostel have taught me to be meticulous and responsible in all that I do.

## **研究目的及意义 Research Purpose and Significance:**

### **Purpose:**

The basic purpose of this research is to synthesis a noble material, more specifically the synthesis of Nano-composite material which has strong nonlinear optical properties.

Nonlinear optics (NLO) is a field of study that explores the interaction of intense light with matter, where the material's response depends nonlinearly on the light's intensity. The primary purpose of NLO research is to understand the fundamental physics behind nonlinear phenomena, such as harmonic generation, self-phase modulation, and optical solitons, and to develop advanced materials with enhanced nonlinear properties, such as high nonlinear susceptibility and ultrafast response times. This research is critical for enabling cutting-edge technologies, including ultrafast lasers, high-speed optical communication systems, and quantum technologies, which rely on nonlinear processes like entangled photon generation. Beyond technological innovation, NLO drives scientific advancements by uncovering novel light-matter interactions and fostering the discovery of new materials, such as Perovskites and 2D materials, with tailored optical properties. Its practical applications are vast, ranging from medical imaging techniques like multi-photon microscopy to defence systems that use optical limiters for laser protection. Additionally, NLO contributes to energy efficiency by enabling low-power photonic devices. As an interdisciplinary field, NLO bridges physics, chemistry, materials science, and engineering, addressing global challenges in communication, healthcare, and information processing. In essence, nonlinear optics research is pivotal for advancing both fundamental science and transformative technologies, making it a cornerstone of modern photonics and quantum systems.

The significance of this research extends beyond the scientific community, promising to impact several sectors by providing scalable solutions for renewable energy and enhancing the capabilities of communication technologies. Moreover, it represents a crucial step towards strengthening the scientific collaboration between China and Pakistan, contributing to the global effort to address energy sustainability and technological advancement. My aspiration is that this doctoral research will not only contribute to the academic literature but also lay the groundwork for future innovations that benefit society at large.

## **科研准备工作概述及科研方法 Scientific Preparations and Research Methods:**

Using a diverse scientific strategy, I plan to conduct my PhD study by first doing a thorough literature analysis to pinpoint knowledge gaps regarding nanocomposites and nanocrystals, especially with regard to nonlinear optical properties. In order to create nanocrystals with certain characteristics, like improved nonlinear absorption and nonlinear refraction, the experimental phase will concentrate on streamlining synthesis procedures. The synthesized nanocrystals' structural and optical properties will be fully examined using advanced characterization techniques such as photoluminescence spectroscopy transmission electron microscopy (TEM) and X-ray diffraction (XRD). In addition, I will investigate methods for fabricating devices that incorporate these nanocrystals into operational optoelectronic devices, assessing their effectiveness and efficiency in practical settings. This research

approach aims to bridge basic gaps by combining theoretical analysis with experimental investigations.

**Timeline:**

|                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|----------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>1st September 2025 to July 2026</b> | <b>Academic Foundation and literature review:</b> <ul style="list-style-type: none"> <li>• Enroll in doctoral coursework to deepen expertise in the research domain.</li> <li>• Perform an in-depth literature review to identify core concepts, methodologies, and gaps in nonlinear optics.</li> <li>• Initiate a comprehensive review paper on nonlinear optical properties and applications, synthesizing insights from seminal papers and scholarly texts.</li> </ul>                                                                                               |
| <b>September 2026 to February 2027</b> | <b>Research Planning &amp; Theoretical Exploration:</b> <ul style="list-style-type: none"> <li>• Refine research objectives through consultations with supervisors and peers.</li> <li>• Validate foundational theories by replicating results from key studies in nonlinear optics.</li> <li>• Investigate light-matter interactions, emphasizing linear/nonlinear optical responses (e.g., nonlinear absorption [NLA], refraction [NLR] and optical limiting [OL]).</li> </ul>                                                                                         |
| <b>February 2027 to July 2027</b>      | <b>International Exposure and Laboratory Work:</b> <ul style="list-style-type: none"> <li>• Attend international conferences to gain a broader perspective and explore potential collaborations.</li> <li>• Connect with international research groups specializing in nonlinear optics and related areas.</li> <li>• Conduct experimental and simulated laboratory work under my advisor's guidance, based on the research plan.</li> <li>• Publish up to five research papers documenting experimental findings and contribute to the scientific community.</li> </ul> |
| <b>September 2027 to January 2029</b>  | <b>Thesis writing, submission and defence:</b> <ul style="list-style-type: none"> <li>• In this phase I will write and submit a comprehensive thesis.</li> <li>• After the successful submission of my thesis, I will defend it.</li> </ul>                                                                                                                                                                                                                                                                                                                              |

**Conclusion:**

China has been a rapidly rising country that has gained international attention in recent years, especially in artificial intelligence. This outstanding development is a result of the Chinese people's continuous commitment and diligence, which have brought their nation unparalleled achievement. China's rapid economic expansion has strengthened my commitment and confirmed my choice. I am happy that my family is completely in favors of my desire to pursue a doctorate in China. I am

motivated to use my academic work to contribute to this dynamic environment through the success stories of Chinese individuals. China is the perfect place for my doctoral degree because of its dynamic culture, rich history, and forward-thinking educational institutions. A shining example of optical engineering brilliance, Harbin Engineering University aligns with my areas of interest.

I would consider it an honors if I could study at **Herbin Engineering University** in the field of optical engineering under the supervision of Professor **Qiuyun Ouyang** and assure you that I will put forward my utmost efforts into this domain. During this academic journey, I hope to make a significant contribution to optical engineering to meet Pakistan's needs. Advanced studies in China will help my beloved country develop by closing the knowledge gap

Therefore, I am hopeful that the selection and admission committees will help me transform my passion into tangible skills, not only by providing a solid foundation of knowledge and hands-on experience but also by emphasizing their practical and real-world applications.

I look forward to a favorable response and the opportunity to become part of Harbin Engineering University's prestigious academic community, where I can cultivate the skills and knowledge essential for a thriving career in research and academia.

Warm Regards

Muhammad Saleem

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|------------------------------------------------------------------------------------------------------------------------|------------------------------|
| <br>申请人签字<br>SIGNATURE OF APPLICANT   | 日期<br>DATE <b>12-02-2025</b> |
| <br>导师签字<br>SIGNATURE OF SUPERVISOR | 日期<br>DATE <b>25-02-2025</b> |